Write your name here

Surname: Master Functional Skills | Other names: Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number: MFS-UK | Candidate Number: L2-FEB18

Mathematics

Level 2



5 – 9 February 2018
Time: 1 hour 30 minutes

Paper Reference
FSM02/01

You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, compasses.

Total Marks
48/48

My signature confirms that I will not discuss the content of the test with anyone until the end of the 5 day test window.

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Instructions

- Use a **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – there may be more space than you need.
- **Calculators may be used.**

Information

- The total mark for this paper is 48.
- The marks for each question are shown in brackets
  – use this as a guide to how much time to spend on each question.
- **You must show clearly how you get your answers because marks will be awarded for your working out.**
- **Check your working and your answers at each stage.**
- **This sign shows where marks will be awarded for showing your check.**

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.

Turn over ▶

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SECTION A: Adventure holiday

Answer all questions in this section.

Write your answers in the spaces provided.

1 Scott is the leader of a group on an adventure holiday to Morocco.

One night on the holiday the lowest temperature in the desert was –5°C.
The warmest temperature on that day was 16°C.

Scott wants to know the difference between these two temperatures.

Work out the difference between the two temperatures.
Show a check of your working. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Identify warmest and lowest desert temperatures *[M1 temps]*

Warmest day temperature = 16°C

Lowest night temperature = –5°C

Step 2: Calculate the difference between the temperatures *[M1 diff]*

Difference = 16 – (–5) = 16 + 5 = 21°C

Alternative number line method:

From –5°C up to 0°C = 5°C; From 0°C up to 16°C = 16°C

Total difference = 5 + 16 = 21°C

Final Answer: 21°C *[A1 answer]*

Use the box below to show your check.

☑

Reverse calculation check: *[B1 check]*

21 – 5 = 16°C (returns warmest temperature)

Or: –5 + 21 = 16°C (verified correct)

(Total for Question 1 is 3 marks)

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2 Scott withdraws 2000 Moroccan dirham from a cash machine in Morocco.

These are the charges that Scott has to pay.

- 2.75% of the cash he withdraws for changing currency.
- 3% of the cash he withdraws for using a cash machine in Morocco.

Scott has to pay both charges in British pounds.

1 British pound = 12.74 Moroccan dirham

What is the total of the charges in British pounds that Scott has to pay?

(4)

Use the box below to show clearly how you get your answer.

Step 1: Calculate currency exchange charge (2.75% of 2000 MAD) *[M1 charge 1]*

Charge 1 = 2000 × 2.75% = 2000 × 0.0275 = 55 Moroccan dirham (MAD)

Step 2: Calculate ATM cash machine charge (3% of 2000 MAD) *[M1 charge 2]*

Charge 2 = 2000 × 3% = 2000 × 0.03 = 60 Moroccan dirham (MAD)

Step 3: Calculate total charges in Moroccan dirham (MAD)

Total charges = 55 + 60 = 115 MAD (or 5.75% of 2000 = 115 MAD)

Step 4: Convert total charges to British pounds (£1 = 12.74 MAD) *[M1 convert]*

Total charges (£) = 115 ÷ 12.74 = £9.02668... ≈ £9.03

Alternative method: Convert withdrawal first to £

2000 ÷ 12.74 = £156.98587...; Total % = 2.75% + 3% = 5.75%

£156.98587 × 0.0575 = £9.02668... ≈ £9.03

Final Answer with correct money notation:

Total Charges = £9.03 (or £9.02) *[A1 answer]*

(Total for Question 2 is 4 marks)

3 Scott has this information about the 18 people in the group.

Gender	Age in years	Number of trips abroad
Female	22	6
Male	38	12
Female	21	4
Male	32	8
Female	46	7
Female	18	3
Male	23	9
Male	28	9
Female	33	11
Male	36	15
Male	27	7
Female	17	2
Male	23	1
Male	25	6
Female	29	8
Female	29	10
Male	31	13
Male	24	7

Scott wants to use a data summary table to sort this information.

He is going to sort the people by

Gender	male, female
Age	under 26, 26 and over
Trips abroad	fewer than 6, 6 or more.

(a) Design and complete a data summary table for Scott.

(5)

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Use the box below to show your answer.

Scott's Group Summary Table (18 People Sorted by Gender, Age & Trips) [M3 design]

Gender	Fewer than 6 trips		6 trips or more		Total
	Under 26	26 & over	Under 26	26 & over	
Male	| (tally) 1	None 0	||| (tally) 3	|||| | 6	10
Female	||| (tally) 3	None 0	| (tally) 1	|||\ 4	8
Total	4	0	4	10	18

[A2 completed]

Data Verification from List of 18 People:

- Male under 26 (<6 trips): 1 person (Age 23, 1 trip)
- Male under 26 (6+ trips): 3 people (Ages 23 [9], 25 [6], 24 [7])
- Male 26+ (<6 trips): 0 people
- Male 26+ (6+ trips): 6 people (Ages 38 [12], 32 [8], 28 [9], 36 [15], 27 [7], 31 [13])
- Female under 26 (<6 trips): 3 people (Ages 21 [4], 18 [3], 17 [2])
- Female under 26 (6+ trips): 1 person (Age 22 [6])
- Female 26+ (<6 trips): 0 people
- Female 26+ (6+ trips): 4 people (Ages 46 [7], 33 [11], 29 [8], 29 [10])

Total Sum Check: 1 + 0 + 3 + 6 + 3 + 0 + 1 + 4 = 18 people. (Complete & verified)

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Abby, Belle and Zak are part of the group.
They are going to fly from Spain to Morocco.

The luggage charges for the flight are

- up to 23 kg free
- 12 euros for every kg over 23 kg up to 32 kg.

Any piece of luggage with a weight of over 32 kg is not allowed on the flight.

Abby has 21 kg of luggage.
Belle has 36 kg of luggage.
Zak has 30 kg of luggage.

They must take all of this luggage.

Abby, Belle and Zak can swap items between their luggage.
They need to know the total cost of taking all the luggage.

(b) Find the total cost of taking all the luggage.
State the weight of luggage each person has after the swap. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total weight and excess baggage charges *[M1 weight]*

Individual weights: Abby = 21 kg, Belle = 36 kg, Zak = 30 kg

Total luggage weight = 21 + 36 + 30 = 87 kg

Total free baggage allowance = 3 passengers × 23 kg = 69 kg

Total excess weight = 87 − 69 = 18 kg

Total excess baggage fee = 18 kg × 12 euros = 216 euros *[A1 cao]*

Step 2: Check airline luggage limits & redistribute weight *[M1 swap]*

Rule: Any luggage over 32 kg is strictly NOT allowed on the flight.

Belle has 36 kg (exceeds 32 kg limit by 4 kg: 36 − 32 = 4 kg).

Swap: Belle transfers 4 kg from her luggage to Abby.

Final luggage weights after swap (Option 1):

- Abby: 21 + 4 = 25 kg (≤ 32 kg limit)
- Belle: 36 − 4 = 32 kg (≤ 32 kg limit)
- Zak: 30 kg (remains unchanged, ≤ 32 kg limit) *[A1 weights]*

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Q3(b) Continuation & Alternative Valid Redistributon:

Alternative Option 2: Equal weight distribution

Total weight = 87 kg ÷ 3 passengers = 29 kg each

- **Abby: 29 kg (transfers 8 kg from Belle)**
- **Belle: 29 kg (transfers 7 kg to Abby, 0 kg to Zak)**
- **Zak: 29 kg (transfers 1 kg to Abby)**

All weights (29 kg) are strictly below the 32 kg maximum limit.

Summary of Final Answers:

- **Total Cost of taking all luggage = 216 euros**
- **Weight after swap: Abby = 25 kg, Belle = 32 kg, Zak = 30 kg**
  (or Abby = 29 kg, Belle = 29 kg, Zak = 29 kg)

(Total for Question 3 is 9 marks)

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SECTION B : Coffee shops

Answer all questions in this section.

Write your answers in the spaces provided.

4 Enid owns some coffee shops.
Her coffee shops are open for 6 days each week.

Enid wants to find the average sales from each of her coffee shops.

The table gives information about sales of coffee from her coffee shop in London for one week.

	Mon	Tue	Wed	Thu	Fri	Sat
Number of cups of coffee sold	225	188	208	241	222	320

(a) Work out the average number of cups of coffee sold per day for Enid.
Show a check of your working. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total number of cups of coffee sold in the week (6 days) *[M1 sum]*

Daily sales: Mon = 225, Tue = 188, Wed = 208,
Thu = 241, Fri = 222, Sat = 320 cups

Total Sales = 225 + 188 + 208 + 241 + 222 + 320 = 1404 cups

Step 2: Calculate mean average sales per day (Total ÷ 6) *[M1 mean]*

Mean = 1404 ÷ 6 = 234 cups of coffee per day

Alternative average (Median check):

Ordered sales: 188, 208, 222, 225, 241, 320

Median = (222 + 225) ÷ 2 = 447 ÷ 2 = 223.5 cups

Final Answer:

Average sales per day = 234 cups (or median 223.5 cups) *[A1 answer]*

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Use the box below to show your check.

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Reverse multiplication check: *[B1 check]*

234 × 6 = 1404 cups (exactly matches weekly total sum)

Alternative check: 1404 − (225 + 188 + 208 + 241 + 222 + 320) = 0 (verified)

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Enid wants to know how the number of sales vary in her coffee shops.
She uses this formula to measure how sales vary.

$$V = \frac{100R}{M}$$

R is the range of the sales
M is the mean of the sales
V is the measure of how sales vary (%)

The table gives information about sales of coffee from her coffee shop in Birmingham for one week.

	Mon	Tue	Wed	Thu	Fri	Sat
Number of cups of coffee sold	190	218	312	280	175	325

The mean number of cups sold in the Birmingham coffee shop is 250
Enid thinks that the value of V for this coffee shop is less than 50%.

(b) Is Enid correct?
Show why you think this. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate the range (R) of sales in Birmingham *[M1 range]*

Highest sales = 325 (Saturday), Lowest sales = 175 (Friday)

Range (R) = 325 − 175 = 150 cups

Step 2: Substitute R = 150 and Mean M = 250 into the formula V = (100 × R) ÷ M *[M1 formula]*

V = (100 × 150) ÷ 250 = 15000 ÷ 250 = 60%

Alternative method: Working backwards from Enid's 50% threshold

If V = 50%, then required Range = (50 × 250) ÷ 100 = 125 cups.

Actual range is 150 cups, which is greater than 125 cups.

Step 3: Compare calculated V with Enid's statement (< 50%)

60% > 50% (the sales variation is 10% higher than Enid thought)

Conclusion: No, Enid is NOT correct. *[A1 decision]*

V = 60%, which is greater than 50% (60% > 50%).

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For her Derby coffee shop Enid uses the formula to work out the value of V using figures for

- one week of sales
- one month of sales.

(c) Which value of V is the more reliable?
Give a reason for your answer. **(1)**

Use the box below to show clearly your answer.

Comparison of sample reliability: *[B1 reason]*

Selected Option: One month of sales is more reliable.

Detailed mathematical justification:

- **Larger sample size: One month contains 4 to 5 times more data points than a single week, giving a far more representative picture of typical sales.**
- **Reduces impact of anomalies: A single week can be easily skewed by unusual weather, local events, or public holidays, whereas a month smooths out variations.**

(Total for Question 4 is 8 marks)

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5 Enid wants to work out the profit one of her coffee shops makes from selling tea and coffee.

She knows that

- on a normal day the coffee shop sells 160 cups of tea
- 3 cups of coffee are sold for every 2 cups of tea sold
- the income from a cup of tea is £1.80
- the income from a cup of coffee is £2.20

The profit from the sale of tea and coffee is $\frac{1}{3}$ of the total income from the sales of cups of tea and cups of coffee.

Work out the total profit from the sales of cups of tea and cups of coffee on a normal day for Enid. **(5)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate number of cups of coffee sold using the ratio 3 : 2 *[M1 ratio]*

Ratio Coffee : Tea = 3 : 2. Tea sold = 160 cups (represents 2 parts)

1 part = 160 ÷ 2 = 80 cups

Coffee sold (3 parts) = 3 × 80 = 240 cups of coffee

Step 2: Calculate total sales income from tea and coffee *[M1 income]*

- Income from Tea = 160 cups × £1.80 = £288.00
- Income from Coffee = 240 cups × £2.20 = £528.00

Total Income = £288.00 + £528.00 = £816.00

Step 3: Calculate profit (1/3 of total income) *[M1 profit]*

Profit = 1/3 × £816.00 = £816.00 ÷ 3 = £272.00

Alternative check by unit profit:

Tea profit: £288 ÷ 3 = £96.00; Coffee profit: £528 ÷ 3 = £176.00

Total Profit = £96.00 + £176.00 = £272.00 (verified)

Final Answer: Total Profit = £272.00 *[A1 answer]*

(Total for Question 5 is 5 marks)

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6 Enid uses part-time staff in her coffee shops.

In her Taunton coffee shop she needs

- 2 part-time staff working from 7 am to 9 am
- 3 part-time staff working from 12 pm to 4 pm
- 1 part-time staff working from 4 pm to 5 pm.

Here is the list of part-time staff and information about when they can work.

Name	Number of hours available	Notes
Leon	2	Morning only
Naomi	4	Starts at 12 o'clock or later
Pat	3	Starts 2 pm
Roma	3	Starts at 12 o'clock or later
Shola	5	Any time

Only one of Roma or Shola is available to work between 3 pm and 4 pm.

Enid needs to organise a schedule for the part-time staff.

Complete the schedule for Enid. **(3)**

Complete the table below.

	7 – 8am	8 – 9am	12 – 1pm	1 – 2pm	2 –3pm	3 – 4pm	4 – 5pm
Leon	☐	☐					
Naomi			☐	☐	☐	☐	
Pat					☐	☐	☐
Roma			☐	☐	☐		
Shola	☐	☐	☐	☐		☐	

(Total for Question 6 is 3 marks)

Schedule Verification & Constraints Check: *[A3 complete rota]*

- **Staff per time slot:**
  7–8am: 2 staff | 8–9am: 2 staff | 12–1pm: 3 staff | 1–2pm: 3 staff
  2–3pm: 3 staff | 3–4pm: 3 staff | 4–5pm: 1 staff (All requirements met!)
- **Individual Hours: Leon (2h morning), Naomi (4h starts 12pm), Pat (3h starts 2pm), Roma (3h starts 12pm), Shola (5h any time). Only Shola works 3–4pm (constraint satisfied)**

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SECTION C : Building

Answer all questions in this section.

Write your answers in the spaces provided.

7 Jose is a builder.
He is going to build a new garage.
The garage will be in the shape of a cuboid.

He has this drawing of the garage.

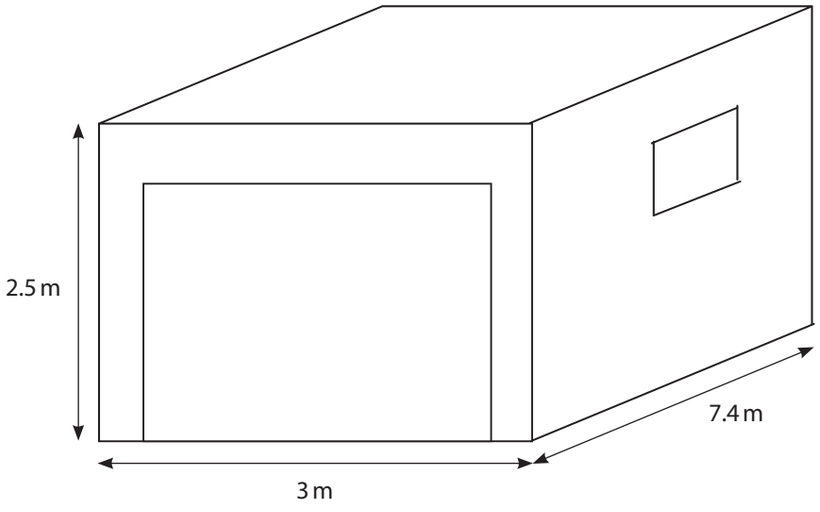


Diagram **not** accurately drawn

Jose decides that

- $\frac{3}{4}$ of the total area of the four sides of the garage will be made of bricks
- he will not use any bricks for the floor or the roof.

Jose knows 1 m^2 of brick wall needs 60 bricks.

He will buy the bricks in packs.
There are 360 bricks in each pack.

Jose thinks that 7 packs of bricks is enough for the garage.

(a) Is Jose correct?
Show why you think this. **(6)**

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Use the box below to show clearly how you get your answer.

Step 1: Calculate total surface area of the four garage walls [M1 wall areas]

Dimensions: Length = 7.4 m, Width = 3.0 m, Height = 2.5 m

- 2 Long walls: 2 × (7.4 m × 2.5 m) = 2 × 18.5 m² = 37.0 m²
- 2 Short walls: 2 × (3.0 m × 2.5 m) = 2 × 7.5 m² = 15.0 m²

Total wall area = 37.0 + 15.0 = 52.0 m²

Step 2: Calculate brick area (3/4 of total wall area) [M1 3/4 area]

Brick wall area = 3/4 × 52.0 m² = 0.75 × 52.0 = 39.0 m²

Step 3: Calculate number of bricks needed (60 bricks per 1 m^2) [M1 bricks needed]

Bricks needed = 39.0 m² × 60 bricks/m² = 2,340 bricks

Step 4: Calculate bricks provided by Jose (7 packs of 360 bricks) [M1 Jose bricks]

Bricks ordered = 7 × 360 = 2,520 bricks

Step 5: Compare figures & conclude (packs needed vs ordered)

2,520 > 2,340 (Jose has ordered 180 extra bricks)

Alternative packs check: 2,340 ÷ 360 = 6.5 packs □ rounds up to 7 packs

Alternative area check: (7 × 360) ÷ 60 = 42 m²; 42 m² > 39 m²

Conclusion: Yes, Jose is correct (2520 bricks ordered > 2340 bricks needed) [A1 decision]

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Jose has to make 1.7 tonnes of dry mortar for the garage.

He uses cement, sand and lime in the ratio 1 : 5 : 2 to make the dry mortar.

Jose wants to know how much lime he needs to make 1.7 tonnes of dry mortar.

(b) How much lime does Jose need?
Give your answer in tonnes correct to 2 decimal places.
Show a check of your working. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total parts in the mortar ratio (Cement : Sand : Lime = 1 : 5 : 2) *[M1 ratio]*

Total parts = 1 + 5 + 2 = 8 equal parts

Step 2: Calculate weight of one part for 1.7 tonnes of mortar

1 part = 1.7 tonnes ÷ 8 = 0.2125 tonnes

Step 3: Calculate weight of lime needed (2 parts) *[M1 lime]*

Lime weight = 2 × 0.2125 = 0.425 tonnes

Step 4: Round answer correct to 2 decimal places

0.425 rounds up to 0.43 tonnes (since the 3rd decimal digit is 5)

Individual ingredient breakdown for full understanding:

Cement (1 part) = 0.2125 t; Sand (5 parts) = 1.0625 t; Lime (2 parts) = 0.425 t

Final Answer: 0.43 tonnes of lime *[A1 answer]*

Use the box below to show your check.

☑

Reverse calculation check: *[B1 check]*

0.425 ÷ 2 × 8 = 0.2125 × 8 = 1.7 tonnes (returns total mortar)

Alternative check: 0.425 is 2/8 = 1/4 of total; 1.7 ÷ 4 = 0.425 t (verified)

(Total for Question 7 is 9 marks)

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8 Jose has to fit a concrete support in a house.

The support is in the shape of a cuboid, 3 m by 0.1 m by 0.2 m.

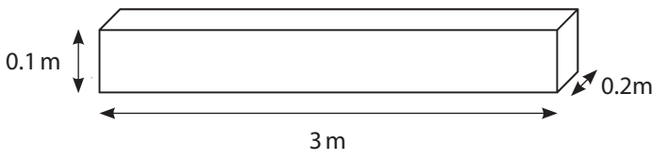


Diagram **not** accurately drawn

Jose wants to work out the weight of the support.

He knows that the weight of 1 m^3 of concrete is 2500 kg.

Jose thinks that the weight of the support will be about 150 kg.

Is Jose correct?
Show why you think this. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate volume of the cuboid support (L × W × H) *[M1 volume]*

Volume = 3 m × 0.2 m × 0.1 m = 0.06 m^3

Step 2: Calculate concrete weight (Volume × 2500 kg/m^3) *[M1 weight]*

Weight = 0.06 m^3 × 2500 kg/m^3 = 150 kg

Step 3: Compare with Jose's estimate (150 kg) & conclude

Conclusion: Yes, Jose is correct (Calculated 150 kg = estimate 150 kg). *[A1 answer]*

(Total for Question 8 is 3 marks)

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9 Jose is asked to make a base for a statue.
The base will be a prism.
The cross section of the prism will be a hexagon.

The hexagon consists of 6 equilateral triangles.
Each side of the hexagon is of length 30 cm.

Jose wants to make a scale drawing of the hexagon.

He has already drawn one side of the hexagon.

(a) Give the scale that Jose has used.

(1)

Write your answer in the box below.

Calculation of drawing scale: *[B1 scale]*

- Length of drawn line in box opposite = 3 cm
- Actual real length of hexagon side = 30 cm
- Scale ratio = 3 cm : 30 cm = 1 cm : 10 cm (or 1 : 10)

Final Answer: 1 cm represents 10 cm (1 : 10)

(b) Complete the scale drawing of the hexagon.
Remember to use the line Jose has drawn.

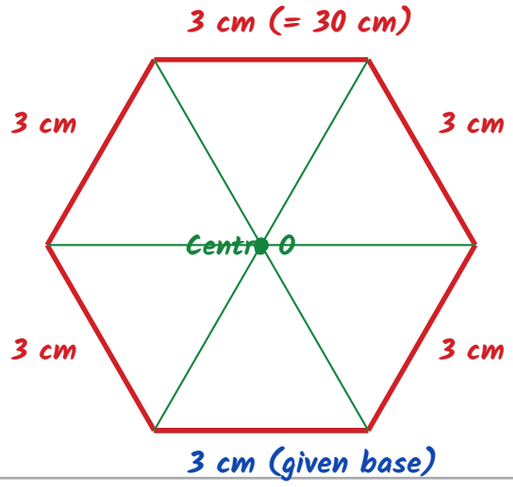
(3)

Use the box opposite to complete the scale drawing.

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Scale Drawing: Regular Hexagon (Composed of 6 Equilateral Triangles) [M3 drawing]

- Scale: 1 cm on drawing represents 10 cm real statue base (1 : 10)
- Each side = 3 cm on drawing (85.0 pt = 30 cm real side length)
- Interior angle between adjacent sides = 120° (equilateral triangles angle = 60° at center)



[A1 complete]

(Total for Question 9 is 4 marks)

TOTAL FOR PAPER IS 48 MARKS

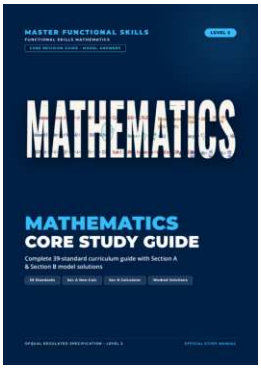
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- Section A Non-Calc & Sec B Calc
- Worked models & golden rules
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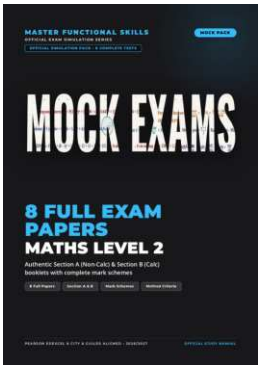


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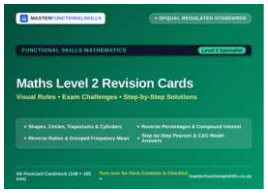


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